

Complete Solutions Manual to Accompany

Contemporary Abstract Algebra

NINTH EDITION

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**CONTEMPORARY ABSTRACT ALGEBRA 9TH EDITION
INSTRUCTOR SOLUTION MANUAL**

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CHAPTER 0

Preliminaries

1. $\{1, 2, 3, 4\}; \{1, 3, 5, 7\}; \{1, 5, 7, 11\}; \{1, 3, 7, 9, 11, 13, 17, 19\};$
 $\{1, 2, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17, 18, 19, 21, 22, 23, 24\}$
2. **a.** 2; 10 **b.** 4; 40 **c.** 4; 120; **d.** 1; 1050 **e.** $pq^2; p^2q^3$
3. 12, 2, 2, 10, 1, 0, 4, 5.
4. $s = -3, t = 2; s = 8, t = -5$
5. By using 0 as an exponent if necessary, we may write $a = p_1^{m_1} \cdots p_k^{m_k}$ and $b = p_1^{n_1} \cdots p_k^{n_k}$, where the p 's are distinct primes and the m 's and n 's are nonnegative. Then $\text{lcm}(a, b) = p_1^{s_1} \cdots p_k^{s_k}$, where $s_i = \max(m_i, n_i)$ and $\text{gcd}(a, b) = p_1^{t_1} \cdots p_k^{t_k}$, where $t_i = \min(m_i, n_i)$. Then $\text{lcm}(a, b) \cdot \text{gcd}(a, b) = p_1^{m_1+n_1} \cdots p_k^{m_k+n_k} = ab$.
6. The first part follows from the Fundamental Theorem of Arithmetic; for the second part, take $a = 4, b = 6, c = 12$.
7. Write $a = nq_1 + r_1$ and $b = nq_2 + r_2$, where $0 \leq r_1, r_2 < n$. We may assume that $r_1 \geq r_2$. Then $a - b = n(q_1 - q_2) + (r_1 - r_2)$, where $r_1 - r_2 \geq 0$. If $a \bmod n = b \bmod n$, then $r_1 = r_2$ and n divides $a - b$. If n divides $a - b$, then by the uniqueness of the remainder, we then have $r_1 - r_2 = 0$. Thus, $r_1 = r_2$ and therefore $a \bmod n = b \bmod n$.
8. Write $as + bt = d$. Then $a's + b't = (a/d)s + (b/d)t = 1$.
9. By Exercise 7, to prove that $(a + b) \bmod n = (a' + b') \bmod n$ and $(ab) \bmod n = (a'b') \bmod n$ it suffices to show that n divides $(a + b) - (a' + b')$ and $ab - a'b'$. Since n divides both $a - a'$ and n divides $b - b'$, it divides their difference. Because $a = a' \bmod n$ and $b = b' \bmod n$ there are integers s and t such that $a = a' + ns$ and $b = b' + nt$. Thus $ab = (a' + ns)(b' + nt) = a'b' + nsb' + a'nt + nsnt$. Thus, $ab - a'b'$ is divisible by n .
10. Write $d = au + bv$. Since t divides both a and b , it divides d . Write $s = mq + r$ where $0 \leq r < m$. Then $r = s - mq$ is a common multiple of both a and b so $r = 0$.
11. Suppose that there is an integer n such that $ab \bmod n = 1$. Then there is an integer q such that $ab - nq = 1$. Since d divides both a and n , d also divides 1. So, $d = 1$. On the other hand, if $d = 1$, then by the corollary of Theorem 0.2, there are integers s and t such that $as + nt = 1$. Thus, modulo n , $as = 1$.

12. $7(5n + 3) - 5(7n + 4) = 1$
13. By the GCD Theorem there are integers s and t such that $ms + nt = 1$.
Then $m(sr) + n(tr) = r$.
14. It suffices to show that $(p^2 + q^2 + r^2) \bmod 3 = 0$. Notice that for any integer a not divisible by 3, $a \bmod 3$ is 1 or 2 and therefore $a^2 \bmod 3 = 1$. So, $(p^2 + q^2 + r^2) \bmod 3 = p^2 \bmod 3 + q^2 \bmod 3 + r^2 \bmod 3 = 3 \bmod 3 = 0$.
15. Let p be a prime greater than 3. By the Division Algorithm, we can write p in the form $6n + r$, where r satisfies $0 \leq r < 6$. Now observe that $6n, 6n + 2, 6n + 3$, and $6n + 4$ are not prime.
16. By properties of modular arithmetic we have
 $(7^{1000}) \bmod 6 = (7 \bmod 6)^{1000} = 1^{1000} = 1$. Similarly,
 $(6^{1001}) \bmod 7 = (6 \bmod 7)^{1001} = -1^{1001} \bmod 7 = -1 = 6 \bmod 7$.
17. Since st divides $a - b$, both s and t divide $a - b$. The converse is true when $\gcd(s, t) = 1$.
18. Observe that $8^{402} \bmod 5 = 3^{402} \bmod 5$ and $3^4 \bmod 5 = 1$. Thus, $8^{402} \bmod 5 = (3^4)^{100} 3^2 \bmod 5 = 4$.
19. If $\gcd(a, bc) = 1$, then there is no prime that divides both a and bc . By Euclid's Lemma and unique factorization, this means that there is no prime that divides both a and b or both a and c . Conversely, if no prime divides both a and b or both a and c , then by Euclid's Lemma, no prime divides both a and bc .
20. If one of the primes did divide $k = p_1 p_2 \cdots p_n + 1$, it would also divide 1.
21. Suppose that there are only a finite number of primes $p_1, p_2, \dots, p_n$. Then, by Exercise 20, $p_1 p_2 \cdots p_n + 1$ is not divisible by any prime. This means that $p_1 p_2 \cdots p_n + 1$, which is larger than any of $p_1, p_2, \dots, p_n$, is itself prime. This contradicts the assumption that $p_1, p_2, \dots, p_n$ is the list of all primes.
22. $\frac{-7}{58} + \frac{3}{58}i$
23. $\frac{-5+2i}{4-5i} = \frac{-5+2i}{4-5i} \frac{4+5i}{4+5i} = \frac{-30}{41} + \frac{-17}{41}i$
24. Let $z_1 = a + bi$ and $z_2 = c + di$. Then $z_1 z_2 = (ac - bd) + (ad + bc)i$; $|z_1| = \sqrt{a^2 + b^2}$, $|z_2| = \sqrt{c^2 + d^2}$, $|z_1 z_2| = \sqrt{a^2 c^2 + b^2 d^2 + a^2 d^2 + b^2 c^2} = |z_1| |z_2|$.
25. x NAND y is 1 if and only if both inputs are 0; x XNOR y is 1 if and only if both inputs are the same.
26. If $x = 1$, the output is y , else it is z .

27. Let S be a set with $n + 1$ elements and pick some a in S . By induction, S has 2^n subsets that do not contain a . But there is one-to-one correspondence between the subsets of S that do not contain a and those that do. So, there are $2 \cdot 2^n = 2^{n+1}$ subsets in all.
28. Use induction and note that

$$2^{n+1}3^{2n+2} - 1 = 18(2^n3^{2n}) - 1 = 18(2^n3^{3n} - 1) + 17.$$
29. Consider $n = 200! + 2$. Then 2 divides n , 3 divides $n + 1$, 4 divides $n + 2, \dots$, and 202 divides $n + 200$.
30. Use induction on n .
31. Say $p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$, where the p 's and the q 's are primes. By the Generalized Euclid's Lemma, p_1 divides some q_i , say q_1 (we may relabel the q 's if necessary). Then $p_1 = q_1$ and $p_2 \cdots p_r = q_2 \cdots q_s$. Repeating this argument at each step we obtain $p_2 = q_2, \dots, p_r = q_r$ and $r = s$.
32. 47. Mimic Example 12.
33. Suppose that S is a set that contains a and whenever $n \geq a$ belongs to S , then $n + 1 \in S$. We must prove that S contains all integers greater than or equal to a . Let T be the set of all integers greater than a that are not in S and suppose that T is not empty. Let b be the smallest integer in T (if T has no negative integers, b exists because of the Well Ordering Principle; if T has negative integers, it can have only a finite number of them so that there is a smallest one). Then $b - 1 \in S$, and therefore $b = (b - 1) + 1 \in S$. This contradicts our assumption that b is not in S .
34. By the Second Principle of Mathematical Induction,

$$f_n = f_{n-1} + f_{n-2} < 2^{n-1} + 2^{n-2} = 2^{n-2}(2 + 1) < 2^n.$$
35. For $n = 1$, observe that $1^3 + 2^3 + 3^3 = 36$. Assume that

$$n^3 + (n + 1)^3 + (n + 2)^3 = 9m \text{ for some integer } m.$$
We must prove that $(n + 1)^3 + (n + 2)^3 + (n + 3)^3$ is a multiple of 9. Using the induction hypothesis we have that

$$(n + 1)^3 + (n + 2)^3 + (n + 3)^3 = 9m - n^3 + (n + 3)^3 = 9m - n^3 + n^3 + 3 \cdot n^2 \cdot 3 + 3 \cdot n \cdot 9 + 3^3 = 9m + 9n^2 + 27n + 27 = 9(m + n^2 + 3n + 3).$$
36. You must verify the cases $n = 1$ and $n = 2$. This situation arises in cases where the arguments that the statement is true for n implies that it is true for $n + 2$ is different when n is even and when n is odd.
37. The statement is true for any divisor of $8^3 - 4 = 508$.
38. One need only verify the equation for $n = 0, 1, 2, 3, 4, 5$. Alternatively, observe that $n^3 - n = n(n - 1)(n + 1)$.
39. Since $3736 \bmod 24 = 16$, it would be 6 p.m.

40. 5

41. Observe that the number with the decimal representation $a_9a_8 \dots a_1a_0$ is $a_910^9 + a_810^8 + \dots + a_110 + a_0$. From Exercise 9 and the fact that $a_i10^i \bmod 9 = a_i \bmod 9$ we deduce that the check digit is $(a_9 + a_8 + \dots + a_1 + a_0) \bmod 9$. So, substituting 0 for 9 or vice versa for any a_i does not change the value of $(a_9 + a_8 + \dots + a_1 + a_0) \bmod 9$.

42. No

43. For the case in which the check digit is not involved, the argument given Exercise 41 applies to transposition errors. Denote the money order number by $a_9a_8 \dots a_1a_0c$ where c is the check digit. For a transposition involving the check digit $c = (a_9 + a_8 + \dots + a_0) \bmod 9$ to go undetected, we must have $a_0 = (a_9 + a_8 + \dots + a_1 + c) \bmod 9$. Substituting for c yields $2(a_9 + a_8 + \dots + a_0) \bmod 9 = a_0$. Then cancelling the a_0 , multiplying by sides by 5, and reducing module 9, we have $10(a_9 + a_8 + \dots + a_1) = a_9 + a_8 + \dots + a_1 = 0$. It follows that $c = a_9 + a_8 + \dots + a_1 + a_0 = a_0$. In this case the transposition does not yield an error.

44. 4

45. Say the number is $a_8a_7 \dots a_1a_0 = a_810^8 + a_710^7 + \dots + a_110 + a_0$. Then the error is undetected if and only if $(a_i10^i - a'_i10^i) \bmod 7 = 0$. Multiplying both sides by 5^i and noting that $50 \bmod 7 = 1$, we obtain $(a_i - a'_i) \bmod 7 = 0$.

46. All except those involving a and b with $|a - b| = 7$.

47. 4

48. Observe that for any integer k between 0 and 8, $k \div 9 = .kkk\dots$

50. 7

51. Say that the weight for a is i . Then an error is undetected if modulo 11, $ai + b(i - 1) + c(i - 2) = bi + c(i - 1) + a(i - 2)$. This reduces to the cases where $(2a - b - c) \bmod 11 = 0$.

52. Say the valid number is $a_1a_2 \dots a_{10}$ and a_i and a_{i+1} were transposed. Then, modulo 11, $10a_1 + 9a_2 + \dots + a_{10} = 0$ and $10a_1 + \dots + (11 - i)a_{i+1} + (11 - (i + 1))a_i + \dots + a_{10} = 5$. Thus, $5 = 5 - 0 = (10a_1 + \dots + (11 - i)a_{i+1} + (11 - (i + 1))a_i + a_{10}) - (10a_1 + 9a_2 + \dots + a_{10})$. It follows that $(a_{i+1} - a_i) \bmod 11 = 5$. Now look for adjacent digits x and y in the invalid number so that $(x - y) \bmod 11 = 5$. Since the only pair is 39, the correct number is 0-669-09325-4.

53. Since $10a_1 + 9a_2 + \cdots + a_{10} = 0 \pmod{11}$ if and only if
 $0 = (-10a_1 - 9a_2 - \cdots - 10a_{10}) \pmod{11} = (a_1 + 2a_2 + \cdots + 10a_{10}) \pmod{11}$,
the check digit would be the same.
54. 7344586061
55. First note that the sum of the digits modulo 11 is 2. So, some digit is 2 too large. Say the error is in position i . Then
 $10 = (4, 3, 0, 2, 5, 1, 1, 5, 6, 8) \cdot (1, 2, 3, 4, 5, 6, 7, 8, 9, 10) \pmod{11} = 2i$. Thus,
the digit in position 5 to 2 too large. So, the correct number is 4302311568.
56. An error in an even numbered position changes the value of the sum by an even amount. However,
 $(9 \cdot 1 + 8 \cdot 4 + 7 \cdot 9 + 6 \cdot 1 + 5 \cdot 0 + 4 \cdot 5 + 3 \cdot 2 + 2 \cdot 6 + 7) \pmod{10} = 5$.
57. 2. Since β is one-to-one, $\beta(\alpha(a_1)) = \beta(\alpha(a_2))$ implies that $\alpha(a_1) = \alpha(a_2)$
and since α is one-to-one, $a_1 = a_2$.
3. Let $c \in C$. There is a b in B such that $\beta(b) = c$ and an a in A such that
 $\alpha(a) = b$. Thus, $(\beta\alpha)(a) = \beta(\alpha(a)) = \beta(b) = c$.
4. Since α is one-to-one and onto we may define $\alpha^{-1}(x) = y$ if and only if
 $\alpha(y) = x$. Then $\alpha^{-1}(\alpha(a)) = a$ and $\alpha(\alpha^{-1}(b)) = b$.
58. $a - a = 0$; if $a - b$ is an integer k then $b - a$ is the integer $-k$; if $a - b$ is
the integer n and $b - c$ is the integer m , then $a - c = (a - b) + (b - c)$ is
the integer $n + m$. The set of equivalence classes is
 $\{[k] \mid 0 \leq k < 1, \ k \text{ is real}\}$. The equivalence classes can be represented by
the real numbers in the interval $[0, 1)$. For any real number a , $[a] = \{a + k \mid$
where k ranges over all integers $\}$.
59. No. $(1, 0) \in R$ and $(0, -1) \in R$ but $(1, -1) \notin R$.
60. Obviously, $a + a = 2a$ is even and $a + b$ is even implies $b + a$ is even. If
 $a + b$ and $b + c$ are even, then $a + c = (a + b) + (b + c) - 2b$ is also even. The
equivalence classes are the set of even integers and the set of odd integers.
61. a belongs to the same subset as a . If a and b belong to the subset A and b
and c belong to the subset B , then $A = B$, since the distinct subsets of P
are disjoint. So, a and c belong to A .
62. Suppose that n is odd prime greater than 3 and $n + 2$ and $n + 4$ are also
prime. Then $n \pmod{3} = 1$ or $n \pmod{3} = 2$. If $n \pmod{3} = 1$ then
 $n + 2 \pmod{3} = 0$ and so is not prime. If $n \pmod{3} = 2$ then $n + 4 \pmod{3} = 0$
and so is not prime.

63. The last digit of 3^{100} is the value of $3^{100} \bmod 10$. Observe that $3^{100} \bmod 10$ is the same as $((3^4 \bmod 10)^{25} \bmod 10$ and $3^4 \bmod 10 = 1$. Similarly, the last digit of 2^{100} is the value of $2^{100} \bmod 10$. Observe that $2^5 \bmod 10 = 2$ so that $2^{100} \bmod 10$ is the same as $(2^5 \bmod 10)^{20} \bmod 10 = 2^{20} \bmod 10 = (2^5)^4 \bmod 10 = 2^4 \bmod 10 = 6$.
64. Suppose that there are integers a, b, c , and d with $\gcd(a, b) = 1$ and $\gcd(c, d) = 1$ such that $a^2/b^2 - c^2/d^2 = 1002$. Then $a^2d^2 - c^2b^2 = 1002b^2d^2$. If both b and d are odd, then modulo 4, $b^2 = d^2 = 1$ and $a^2/b^2 - c^2/d^2 = 1002$ reduces to $a^2 - c^2 = 2$. This case is handled in Example 7. If 2^i ($i > 0$) divides b , then a is odd and $a^2d^2 - c^2b^2 = 1002b^2d^2$ implies that 2^i divides d also. It follows that if 2^n is the highest power of 2 that divides one of b or d , then 2^n is the highest power of 2 that divides the other. So dividing both sides of $a^2d^2 - c^2b^2 = 1002b^2d^2$ by 2^n we get an equation of the same form where both b and d are odd. Taking both sides modulo 4 and recalling that for odd x , $x^2 \bmod 4 = 1$ we have that $a^2d^2 - c^2b^2 = 1002b^2d^2$ reduces to $a^2 - c^2 = 2$, which was done in Example 7.
65. Apply γ^{-1} to both sides of $\alpha\gamma = \beta\gamma$.

CHAPTER 1

Introduction to Groups

1. Three rotations: 0° , 120° , 240° , and three reflections across lines from vertices to midpoints of opposite sides.
2. Let $R = R_{120}$, $R^2 = R_{240}$, F a reflection across a vertical axis, $F' = RF$ and $F'' = R^2F$

	R_0	R	R^2	F	F'	F''
R_0	R_0	R	R^2	F	F'	F''
R	R	R^2	R_0	F'	F''	F
R^2	R^2	R_0	R	F''	F	F'
F	F	F''	F'	R_0	R^2	R
F'	F'	F	F''	R	R_0	R^2
F''	F''	F'	F	R^2	R	R_0

3. **a.** V **b.** R_{270} **c.** R_0 **d.** $R_0, R_{180}, H, V, D, D'$ **e.** none
4. Five rotations: 0° , 72° , 144° , 216° , 288° , and five reflections across lines from vertices to midpoints of opposite sides.
5. D_n has n rotations of the form $k(360^\circ/n)$, where $k = 0, \dots, n-1$. In addition, D_n has n reflections. When n is odd, the axes of reflection are the lines from the vertices to the midpoints of the opposite sides. When n is even, half of the axes of reflection are obtained by joining opposite vertices; the other half, by joining midpoints of opposite sides.
6. A nonidentity rotation leaves only one point fixed – the center of rotation. A reflection leaves the axis of reflection fixed. A reflection followed by a different reflection would leave only one point fixed (the intersection of the two axes of reflection) so it must be a rotation.
7. A rotation followed by a rotation either fixes every point (and so is the identity) or fixes only the center of rotation. However, a reflection fixes a line.
8. In either case, the set of points fixed is some axis of reflection.
9. Observe that $1 \cdot 1 = 1$; $1(-1) = -1$; $(-1)1 = -1$; $(-1)(-1) = 1$. These relationships also hold when 1 is replaced by a “rotation” and -1 is replaced by a “reflection.”
10. reflection.

11. Thinking geometrically and observing that even powers of elements of a dihedral group do not change orientation we note that each of a, b and c appears an even number of times in the expression. So, there is no change in orientation. Thus, the expression is a rotation. Alternatively, as in Exercise 9, we associate each of a, b and c with 1 if they are rotations and -1 if they are reflections and we observe that in the product $a^2b^4ac^5a^3c$ the terms involving a represents six 1s or six -1 s, the term b^4 represents four 1s or four -1 s, and the terms involving c represents six 1s or six -1 s. Thus the product of all the 1s and -1 s is 1. So the expression is a rotation.
12. H, I, O, X. Rotations of $0^\circ, 180^\circ$, horizontal reflection, and vertical reflection.
13. In D_4 , $HD = DV$ but $H \neq V$.
14. D_n is not commutative.
15. R_0, R_{180}, H, V
16. Rotations of 0° and 180° ; Rotations of 0° and 180° and reflections about the diagonals.
17. R_0, R_{180}, H, V
18. Let the distance from a point on one H to the corresponding point on an adjacent H be one unit. Then translations of any number of units to the right or left are symmetries; reflection across the horizontal axis through the middle of the H 's is a symmetry; reflection across any vertical axis midway between two H 's or bisecting any H is a symmetry. All other symmetries are compositions of finitely many of those already described. The group is non-Abelian.
19. In each case the group is D_6 .
20. D_{28}
21. First observe that $X^2 \neq R_0$. Since R_0 and R_{180} are the only elements in D_4 that are squares we have $X^2 = R_{180}$. Solving $X^2Y = R_{90}$ for Y gives $Y = R_{270}$.
22. $X^2 = F$ has no solutions; the only solution to $X^3 = F$ is F .
23. 180° rotational symmetry.
24. Z_4, D_5, D_4, Z_2
 D_4, Z_3, D_3, D_{16}
 D_7, D_4, D_5, Z_{10}
25. Their only symmetry is the identity.

CHAPTER 2

Groups

1. **c, d**
2. **c, d**
3. none
4. **a, c**
5. $7; 13; n-1; \frac{1}{3-2i} = \frac{1}{3-2i} \frac{3+2i}{3+2i} = \frac{3}{13} + \frac{2}{13}i$
6. **a.** $-31-i$ **b.** 5 **c.** $\frac{1}{12} \begin{bmatrix} 2 & -3 \\ -8 & 6 \end{bmatrix}$ **d.** $\begin{bmatrix} 2 & 4 \\ 4 & 6 \end{bmatrix}$.
7. The set does not contain the identity; closure fails.
8. $1, 3, 7, 9, 11, 13, 17, 19$.
9. Under multiplication modulo 4, 2 does not have an inverse. Under multiplication modulo 5, $\{1, 2, 3, 4\}$ is closed, 1 is the identity, 1 and 4 are their own inverses, and 2 and 3 are inverses of each other. Modulo multiplication is associative.
10. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.
11. a^{11}, a^6, a^4, a^1
12. $5, 4, 8$
13. (a) $2a + 3b$; (b) $-2a + 2(-b + c)$; (c) $-3(a + 2b) + 2c = 0$
14. $(ab)^3 = ababab$ and $(ab^{-2}c)^{-2} = ((ab^{-2}c)^{-1})^2 = (c^{-1}b^2a^{-1})^2 = c^{-1}b^2a^{-1}c^{-1}b^2a^{-1}$.
15. Observe that $a^5 = e$ implies that $a^{-2} = a^3$ and $b^7 = e$ implies that $b^{14} = e$ and therefore $b^{-11} = b^3$. Thus, $a^{-2}b^{-11} = a^3b^3$. Moreover, $(a^2b^4)^{-2} = ((a^2b^4)^{-1})^2 = (b^{-4}a^{-2})^2 = (b^3a^3)^2$.
16. The identity is 25.
17. Since the inverse of an element in G is in G , $H \subseteq G$. Let g belong to G . Then g^{-1} belongs to G and therefore $(g^{-1})^{-1} = g$ belong to G . So, $G \subseteq H$.
18. $K = \{R_0, R_{180}\}; L = \{R_0, R_{180}, H, V, D, D'\}$.

19. The set is closed because $\det(AB) = (\det A)(\det B)$. Matrix multiplication is associative. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is the identity.
- Since $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ its determinant is $ad - bc = 1$.
20. $1^2 = (n-1)^2 = 1$.
21. Using closure and trial and error, we discover that $9 \cdot 74 = 29$ and 29 is not on the list.
22. Consider $xyx = xyx$.
23. For $n \geq 0$, we use induction. The case that $n = 0$ is trivial. Then note that $(ab)^{n+1} = (ab)^n ab = a^n b^n ab = a^{n+1} b^{n+1}$. For $n < 0$, note that $e = (ab)^0 = (ab)^n (ab)^{-n} = (ab)^n a^{-n} b^{-n}$ so that $a^n b^n = (ab)^n$. In a non-Abelian group $(ab)^n$ need not equal $a^n b^n$.
24. The “inverse” of putting on your socks and then putting on your shoes is taking off your shoes then taking off your socks. Use D_4 for the examples. (An appropriate name for the property $(abc)^{-1} = c^{-1}b^{-1}a^{-1}$ is “Socks-Shoes-Boots Property.”)
25. Suppose that G is Abelian. Then by Exercise 24, $(ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$. If $(ab)^{-1} = a^{-1}b^{-1}$ then by Exercise 24 $e = aba^{-1}b^{-1}$. Multiplying both sides on the right by ba yields $ba = ab$.
26. By definition, $a^{-1}(a^{-1})^{-1} = e$. Now multiply on the left by a .
27. The case where $n = 0$ is trivial. For $n > 0$, note that $(a^{-1}ba)^n = (a^{-1}ba)(a^{-1}ba) \cdots (a^{-1}ba)$ (n terms). So, cancelling the consecutive a and a^{-1} terms gives $a^{-1}b^n a$. For $n < 0$, note that $e = (a^{-1}ba)^n (a^{-1}ba)^{-n} = (a^{-1}ba)^n (a^{-1}b^{-n}a)$ and solve for $(a^{-1}ba)^n$.
28. $(a_1 a_2 \cdots a_n)(a_n^{-1} a_{n-1}^{-1} \cdots a_2^{-1} a_1^{-1}) = e$
29. By closure we have $\{1, 3, 5, 9, 13, 15, 19, 23, 25, 27, 39, 45\}$.
30. Z_{105} ; Z_{44} and D_{22} .
31. Suppose x appears in a row labeled with a twice. Say $x = ab$ and $x = ac$. Then cancellation gives $b = c$. But we use distinct elements to label the columns.
- 32.
- | | 1 | 5 | 7 | 11 |
|----|----|----|----|----|
| 1 | 1 | 5 | 7 | 11 |
| 5 | 5 | 1 | 11 | 7 |
| 7 | 7 | 11 | 1 | 5 |
| 11 | 11 | 7 | 5 | 1 |

33. Proceed as follows. By definition of the identity, we may complete the first row and column. Then complete row 3 and column 5 by using Exercise 31. In row 2 only c and d remain to be used. We cannot use d in position 3 in row 2 because there would then be two d 's in column 3. This observation allows us to complete row 2. Then rows 3 and 4 may be completed by inserting the unused two elements. Finally, we complete the bottom row by inserting the unused column elements.
34. $(ab)^2 = a^2b^2 \Leftrightarrow abab = aabb \Leftrightarrow ba = ab$.
 $(ab)^{-2} = b^{-2}a^{-2} \Leftrightarrow b^{-1}a^{-1}b^{-1}a^{-1} = b^{-1}b^{-1}a^{-1}a^{-1} \Leftrightarrow a^{-1}b^{-1} = b^{-1}a^{-1} \Leftrightarrow ba = ab$.
35. $axb = c$ implies that $x = a^{-1}(axb)b^{-1} = a^{-1}cb^{-1}$; $a^{-1}xa = c$ implies that $x = a(a^{-1}xa)a^{-1} = aca^{-1}$.
36. Observe that $xabx^{-1} = ba$ is equivalent to $xab = bax$ and this is true for $x = b$.
37. Since e is one solution it suffices to show that nonidentity solutions come in distinct pairs. To this end note that if $x^3 = e$ and $x \neq e$, then $(x^{-1})^3 = e$ and $x \neq x^{-1}$. So if we can find one nonidentity solution we can find a second one. Now suppose that a and a^{-1} are nonidentity elements that satisfy $x^3 = e$ and b is a nonidentity element such that $b \neq a$ and $b \neq a^{-1}$ and $b^3 = e$. Then, as before, $(b^{-1})^3 = e$ and $b \neq b^{-1}$. Moreover, $b^{-1} \neq a$ and $b^{-1} \neq a^{-1}$. Thus, finding a third nonidentity solution gives a fourth one. Continuing in this fashion we see that we always have an even number of nonidentity solutions to the equation $x^3 = e$.
 To prove the second statement note that if $x^2 \neq e$, then $x^{-1} \neq x$ and $(x^{-1})^2 \neq e$. So, arguing as in the preceding case we see that solutions to $x^2 \neq e$ come in distinct pairs.
38. In D_4 , $HR_{90}V = DR_{90}H$ but $HV \neq DH$.
39. Observe that $aa^{-1}b = ba^{-1}a$. Cancelling the middle term a^{-1} on both sides we obtain $ab = ba$.
40. $X = VR_{270}D'H$.
41. If $F_1F_2 = R_0$ then $F_1F_2 = F_1F_1$ and by cancellation $F_1 = F_2$.
42. Observe that $F_1F_2 = F_2F_1$ implies that $(F_1F_2)(F_1F_2) = R_0$. Since F_1 and F_2 are distinct and F_1F_2 is a rotation it must be R_{180} .
43. Since FR^k is a reflection we have $(FR^k)(FR^k) = R_0$. Multiplying on the left by F gives $R^kFR^k = F$.
44. Since FR^k is a reflection we have $(FR^k)(FR^k) = R_0$. Multiplying on the right by R^{-k} gives $FR^kF = R^{-k}$. If D_n were Abelian, then $FR_{360^\circ/n}F = R_{360^\circ/n}$. But $(R_{360^\circ/n})^{-1} = R_{360^\circ(n-1)/n} \neq R_{360^\circ/n}$ when $n \geq 3$.

45. **a.** R^3 **b.** R **c.** R^5F
46. Closure and associativity follow from the definition of multiplication; $a = b = c = 0$ gives the identity; we may find inverses by solving the equations $a + a' = 0$, $b' + ac' + b = 0$, $c' + c = 0$ for a', b', c' .
47. Since $a^2 = b^2 = (ab)^2 = e$, we have $aabb = abab$. Now cancel on left and right.
48. If a satisfies $x^5 = e$ and $a \neq e$, then so does a^2, a^3, a^4 . Now, using cancellation we have that a^2, a^3, a^4 are not the identity and are distinct from each other and distinct from a . If these are all of the nonidentity solutions of $x^5 = e$ we are done. If b is another solution that is not a power of a , then by the same argument b, b^2, b^3 and b^4 are four distinct nonidentity solutions. We must further show that b^2, b^3 and b^4 are distinct from a, a^2, a^3, a^4 . If $b^2 = a^i$ for some i , then cubing both sides we have $b = b^6 = a^{3i}$, which is a contradiction. A similar argument applies to b^3 and b^4 . Continuing in this fashion we have that the number of nonidentity solutions to $x^5 = e$ is a multiple of 4. In the general case, the number of solutions is a multiple of 4 or is infinite.
49. The matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is in $\text{GL}(2, Z_2)$ if and only if $ad \neq bc$. This happens when a and d are 1 and at least 1 of b and c is 0 and when b and c are 1 and at least 1 of a and d is 0. So, the elements are
- $$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$
- $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ do not commute.
50. If n is not prime, we can write $n = ab$, where $1 < a < n$ and $1 < b < n$. Then a and b belong to the set $\{1, 2, \dots, n-1\}$ but $0 = ab \bmod n$ does not.
51. Let a be any element in G and write $x = ea$. Then $a^{-1}x = a^{-1}(ea) = (a^{-1}e)a = a^{-1}a = e$. Then solving for x we obtain $x = ae = a$.
52. Suppose that $ab = e$ and let b' be the element in G with the property that $bb' = e$. Then observe that $ba = (ba)e = ba(bb') = b(ab)b' = beb' = (be)b' = bb' = e$.

CHAPTER 3

Finite Groups; Subgroups

1. $|Z_{12}| = 12$; $|U(10)| = 4$; $|U(12)| = 4$; $|U(20)| = 8$; $|D_4| = 8$.
 In Z_{12} , $|0| = 1$; $|1| = |5| = |7| = |11| = 12$; $|2| = |10| = 6$; $|3| = |9| = 4$; $|4| = |8| = 3$; $|6| = 2$.
 In $U(10)$, $|1| = 1$; $|3| = |7| = 4$; $|9| = 2$.
 In $U(20)$, $|1| = 1$; $|3| = |7| = |13| = |17| = 4$; $|9| = |11| = |19| = 2$.
 In D_4 , $|R_0| = 1$; $|R_{90}| = |R_{270}| = 4$; $|R_{180}| = |H| = |V| = |D| = |D'| = 2$.
 In each case, notice that the order of the element divides the order of the group.
2. In Q , $\langle 1/2 \rangle = \{n(1/2) \mid n \in Z\} = \{0, \pm 1/2, \pm 1, \pm 3/2, \dots\}$. In Q^* ,
 $\langle 1/2 \rangle = \{(1/2)^n \mid n \in Z\} = \{1, 1/2, 1/4, 1/8, \dots; 2, 4, 8, \dots\}$.
3. In Q , $|0| = 1$. All other elements have infinite order since
 $x + x + \dots + x = 0$ only when $x = 0$.
4. Suppose $|a| = n$ and $|a^{-1}| = k$. Then $(a^{-1})^n = (a^n)^{-1} = e^{-1} = e$. So
 $k \leq n$. Now reverse the roles of a and a^{-1} to obtain $n \leq k$. The infinite
 case follows from the finite case.
5. In Z_{30} , $2 + 28 = 0$ and $8 + 22 = 0$. So, 2 and 28 are inverses of each other
 and 8 and 22 are inverses of each other. In $U(15)$, $2 \cdot 8 = 1$ and $7 \cdot 13 = 1$.
 So, 2 and 8 are inverses of each other and 7 and 13 are inverses of each
 other.
6. **a.** $|6| = 2$, $|2| = 6$, $|8| = 3$; **b.** $|3| = 4$, $|8| = 5$, $|11| = 12$;
c. $|5| = 12$, $|4| = 3$, $|9| = 4$. In each case $|a + b|$ divides $\text{lcm}(|a|, |b|)$.
7. $(a^4c^{-2}b^4)^{-1} = b^{-4}c^2a^{-4} = b^3c^2a^2$.
8. If a subgroup of D_3 contains R_{240} and F it also contains
 $R_0, R_{240}^2 = R_{120}, R_{240}F$, and $R_{120}F$, which is all six elements of D_3 . If F
 and F' are distinct reflections in a subgroup of D_3 , then $FF' = R_{240}$ is
 also in the subgroup. Thus the subgroup must be D_3 .
9. If a subgroup of D_4 contains R_{270} and a reflection F , then it also contains
 the six other elements $R_0, (R_{270})^2 = R_{180}, (R_{270})^3 = R_{90}, R_{270}F, R_{180}F$
 and $R_{90}F$. If a subgroup of D_4 contains H and D , then it also contains
 $HD = R_{90}$ and $DH = R_{270}$. But this implies that the subgroup contains
 every element of D_4 . If it contains H and V then it contains $HV = R_{180}$
 and R_0 .

10. $\{R_0, R_{90}, R_{180}, R_{270}\}$, $\{R_0, R_{180}, H, V\}$, and $\{R_0, R_{180}, D, D'\}$.
11. If n is a positive integer, the real solutions of $x^n = 1$ are 1 when n is odd and ± 1 when n is even. So, the only elements of finite order in R^* are ± 1 .
12. 1 or 2.
13. By Exercise 27 of Chapter 2 we have $e = (xax^{-1})^n = xa^n x^{-1}$ if and only if $a^n = e$.
14. By Exercise 13, for every x in G $|xax^{-1}| = |a|$, so that $xax^{-1} = a$ or $xa = ax$.
15. Suppose $G = H \cup K$. Pick $h \in H$ with $h \notin K$. Pick $k \in K$ with $k \notin H$. Then, $hk \in G$ but $hk \notin H$ and $hk \notin K$. $U(8) = \{1, 3\} \cup \{1, 5\} \cup \{1, 7\}$.
16. ∞
17. $U_4(20) = \{1, 9, 13, 17\}$; $U_5(20) = \{1, 11\}$; $U_5(30) = \{1, 11\}$;
 $U_{10}(30) = \{1, 11\}$.
 To prove that $U_k(n)$ is a subgroup it suffices to show that it is closed. Suppose that a and b belong to $U_k(n)$. We must show that in $U(n)$, $ab \bmod k = 1$. That is, $(ab \bmod n) \bmod k = 1$. Let $n = kt$ and $ab = qn + r$ where $0 \leq r < n$. Then
 $(ab \bmod n) \bmod k = r \bmod k = (ab - qn) \bmod k = (ab - qkt) \bmod k = ab \bmod k = (a \bmod k)(b \bmod k) = 1 \cdot 1 = 1$. H is not a subgroup because $7 \in H$ but $7 \cdot 7 = 9$ is not $1 \bmod 3$.
18. The possibilities are 1, 2, 3 and 6. 5 is not possible for if $a^5 = e$, then $e = a^6 = aa^5 = a$. 4 is not possible for if $a^4 = e$, then $e = a^6 = a^2 a^4 = a^2$.
19. Suppose that $m < n$ and $a^m = a^n$. Then $e = a^n a^{-m} = a^{n-m}$. This contradicts the assumption that a has infinite order.
20. If both ab and ba have infinite order, we are done. Suppose that $|ab| = n$. Then $e = (ab)(ab) \cdots (ab)$ [n factors]. Thus,
 $ba = bea = b((ab)(ab) \cdots (ab))a = (ba)^{n+1}$. This shows that $(ba)^n = e$ so that $|ba| \leq |ab|$. By symmetry, $|ab| \leq |ba|$.
21. If a has infinite order, then $e, a, a^2, \dots$ are all distinct and belong to G , so G is infinite. If $|a| = n$, then $a^i = a^j$ where $0 < i < j < n$ implies $a^{j-i} = e$, which is a contradiction. Thus, $e, a, a^2, \dots, a^{n-1}$ are all distinct and belong to G , so G has at least n elements.
22. $\langle 3 \rangle = \{3, 3^2, 3^3, 3^4, 3^5, 3^6\} = \{3, 9, 13, 11, 5, 1\} = U(14)$. $\langle 5 \rangle = \{5, 5^2, 5^3, 5^4, 5^5, 5^6\} = \{5, 11, 13, 9, 3, 1\} = U(14)$. $\langle 11 \rangle = \{11, 9, 1\} \neq U(14)$.
23. Since $|U(20)| = 8$, for $U(20) = \langle k \rangle$ for some k it must be the case that $|k| = 8$. But $1^1 = 1$, $3^4 = 1$, $7^4 = 1$, $9^2 = 1$, $11^2 = 1$, $13^4 = 1$, $17^4 = 1$, and $19^2 = 1$. So, the maximum order of any element is 4.

24. Let A be the subset of even members of Z_n and B the subset of odd members of Z_n . If $x \in B$, then $x + A = \{x + a | a \in A\} \subseteq B$, so $|A| \leq |B|$. Also, $x + B = \{x + b | b \in B\} \subseteq A$, so $|B| \leq |A|$.
25. By Exercise 24, either every element of H is even or exactly half are even. Since H has odd order the latter cannot occur.
26. Suppose that K is a subgroup of D_n that has at least one reflection F . Denote the rotations of K by $R_1, R_2, \dots, R_m$. Then $R_1F, R_2F, \dots, R_mF$ are distinct reflections in K . If F' is any reflection in K , then $F'F = R_i$ for some i . But then $F' = R_iF$. Thus, K has exactly m reflections.
27. By Exercise 26, either every element of H is a rotation or exactly half are rotations. Since H has odd order the latter cannot occur.
28. Suppose that a and b are two elements of order 2 that commute. Then $\{e, a, b, ab\}$ is closed and therefore a subgroup.
29. Observe that by Exercise 26 we have that for any reflection F in D_n the set $\{R_0, R_{180}, F, R_{180}F\}$ is a subgroup of order 4.
30. $\langle 2 \rangle$
31. Let $H = \langle k \rangle$ and observe that because $6 = 30 + 30 - 54$ belongs to H we know 6 is a multiple of k . Thus the possibilities for H are $\langle 6 \rangle, \langle 3 \rangle$ and $\langle 2 \rangle$. None of these can be excluded because each contains 12, 30 and 54.
32. By the corollary to Theorem 0.2, $H = Z$.
33. Suppose that H is a subgroup of D_3 of order 4. Since D_3 has only two elements of order 2, H must contain R_{120} or R_{240} . By closure, it follows that H must contain R_0, R_{120} , and R_{240} as well as some reflection F . But then H must also contain the reflection $R_{120}F$.
34. $H \cap K \neq \emptyset$, since $e \in H \cap K$. Now suppose that $x, y \in H \cap K$. Then, since H and K are subgroups, we know $xy^{-1} \in H$ and $xy^{-1} \in K$. That is, $xy^{-1} \in H \cap K$.
35. If $x \in Z(G)$, then $x \in C(a)$ for all a , so $x \in \bigcap_{a \in G} C(a)$. If $x \in \bigcap_{a \in G} C(a)$, then $xa = ax$ for all a in G , so $x \in Z(G)$.
36. Suppose $x \in C(a)$. Then $xa = ax$. So $a^{-1}(xa) = a^{-1}(ax) = x$. Thus, $(a^{-1}x)a = x$ and therefore $a^{-1}x = xa^{-1}$. This shows $x \in C(a^{-1})$. The other half follows by symmetry.
37. The case that $k = 0$ is trivial. Let $x \in C(a)$. If k is positive, then by induction on k , $xa^{k+1} = xaa^k = axa^k = aa^kx = a^{k+1}x$. The case where k is negative now follows from Exercise 34. The statement "If for some integer k , x commutes a^k , then x commutes with a " is false as can be seen in the group D_4 with $x = H$, $a = R_{90}$ and $k = 2$.

38. Since $|e| = 1$, H is nonempty. Assume $a, b \in H$ and let $|a| = m$ and $|b| = n$. Then $(ab)^{mn} = (a^m)^n(b^n)^m = e^n e^m = ee = e$. So, $|ab|$ divides mn . Since mn is odd, so is $|ab|$.
39. In Z_6 , $H = \{0, 1, 3, 5\}$ is not closed.
40. If $a^2 = b^2$ and $a^3 = b^3$, then $a^2a = b^2b$. Now cancel a^2 and b^2 .
41. **a.** First observe that because $\langle S \rangle$ is a subgroup of G containing S , it is a member of the intersection. So, $H \subseteq \langle S \rangle$. On the other hand, since H is a subgroup of G and H contains S , by definition $\langle S \rangle \subseteq H$.
- b.** Let $K = \{s_1^{n_1} s_2^{n_2} \dots s_m^{n_m} \mid m \geq 1, s_i \in S, n_i \in \mathbb{Z}\}$. Then because K satisfies the subgroup test and contains S we have $\langle S \rangle \subseteq K$. On the other hand, if L is any subgroup of G that contains S then L also contains K by closure. Thus, by part a, $H = \langle S \rangle$ contains K .
42. **a.** $\langle 2 \rangle$ **b.** $\langle 1 \rangle$ **c.** $\langle 3 \rangle$ **d.** $\langle \gcd(m, n) \rangle$ **e.** $\langle 3 \rangle$.
43. Since $ea = ae$, $C(a) \neq \emptyset$. Suppose that x and y are in $C(a)$. Then $xa = ax$ and $ya = ay$. Thus,

$$(xy)a = x(ya) = x(ay) = (xa)y = (ax)y = a(xy)$$

and therefore $xy \in C(a)$. Starting with $xa = ax$, we multiply both sides by x^{-1} on the right and left to obtain $x^{-1}xax^{-1} = x^{-1}axx^{-1}$ and so $ax^{-1} = x^{-1}a$. This proves that $x^{-1} \in C(a)$. By the Two-Step Subgroup Test, $C(a)$ is a subgroup of G .

44. Mimic the proof of Theorem 3.5.
45. No. In D_4 , $C(R_{180}) = D_4$. Yes. Elements in the center commute with all elements.
46. The $C(a) \subseteq C(a^3)$ is easy. To prove the other inclusion, observe that $a^6 = a$ so if $x \in C(a^3)$, then

$$\begin{aligned} xa &= xa^6 = x(a^3a^3) = (xa^3)a^3 = (a^3x)a^3 = a^3(xa^3) \\ &= a^3(a^3x) = (a^3a^3)x = a^6x = ax. \end{aligned}$$

For the second part of the exercise, try D_6 .

47. Let $H = \{x \in G \mid x^n = e\}$. Since $e^1 = e$, $H \neq \emptyset$. Now let $a, b \in H$. Then $a^n = e$ and $b^n = e$. So, $(ab)^n = a^n b^n = ee = e$ and therefore $ab \in H$. Starting with $a^n = e$ and taking the inverse of both sides, we get $(a^n)^{-1} = e^{-1}$. This simplifies to $(a^{-1})^n = e$. Thus, $a^{-1} \in H$. By the Two-Step test, H is a subgroup of G . In D_4 , $\{x \mid x^2 = e\} = \{R_0, R_{180}, H, V, D, D'\}$. This set is not closed because $HD = R_{90}$.

48. For any integer $n \geq 3$, observe that the rotation $R_{360/n}$ in D_n has order n . Now in D_n let F be any reflection. Then $F' = R_{360/n}F$ is a reflection in D_n . Also $|F'| = |F| = 2$ and $F'F = R_{360/n}$ has order n .
49. Let G be a group of even order. Observe that for each element x of order greater than 2 x and x^{-1} are distinct elements of the same order. So, because elements of order greater than 2 come in pairs, there is an even number of elements of order greater than 2 (possibly 0). This means that the number of elements of order 1 or 2 is even. Since the identity is the unique element of order 1, it follows that the number of order 2 is odd.
50. $|A| = 4$, $|B| = 3$, $|AB| = \infty$.
51. First observe that $(a^d)^{n/d} = a^n = e$, so $|a^d|$ is at most n/d . Moreover, there is no positive integer $t < n/d$ such that $(a^d)^t = a^{dt} = e$, for otherwise $|a| \neq n$.
52. In Exercise 50, let $a = A^{-1}$, $b = AB$. Then $ab = A^{-1}AB = B$ has finite order.
53. By induction we will prove that any positive integer n we have

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}.$$

The $n = 1$ case is true by definition. Now assume

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}.$$

Then

$$\begin{aligned} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^{k+1} &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^k \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \\ &= \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & k+1 \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

So, when the entries are from $\mathbf{R}$, $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has infinite order. When the entries are from Z_p , the order is p .

54. For the first part use induction; 6, ∞ .
55. For any positive integer n , a rotation of $360^\circ/n$ has order n . If we let R be a rotation of $\sqrt{2}$ degrees then R^n is a rotation of $\sqrt{2}n$ degrees. This is never a multiple of 360° , for if $\sqrt{2}n = 360k$ then $\sqrt{2} = 360k/n$, which is rational. So, R has infinite order.
56. Let $a = 2$ and $b = -2^{-1}$.

57. Inscribe a regular n -gon in a circle. Then every element of D_n is a symmetry of the circle.
58. If $|a| = m$ and $|b| = n$, then $(ab^{-1})^{mn} = e$. The elements of finite order do not always form a subgroup in a non-Abelian group. (See Exercise 50 of this set.)
59. Let $|g| = m$ and write $m = nq + r$ where $0 \leq r < n$. Then $g^r = g^{m-nq} = g^m(g^n)^{-q}$ belongs to H . So, $r = 0$.
60. **a.** 2, 2, 4 **b.** 4, 6, 24 **c.** 2, 4, 8 **d.** 2, 4, 8.
61. $1 \in H$, so $H \neq \emptyset$. Let $a, b \in H$. Then $(ab^{-1})^2 = a^2(b^2)^{-1}$, which is the product of two rationals. The integer 2 can be replaced by any positive integer.
62. 2, 4, 16. $|U(rs)| = |U(r)||U(s)|$ when $\gcd(r, s) = 1$
63. $\{1, 9, 11, 19\}$
64. For every nonidentity element a of odd order, a^{-1} is distinct from a and has the same order as a . Thus nonidentity elements of odd order come in pairs. So, there must be some element a of even order, say $|a| = 2m$. Then $|a^m| = 2$.
65. Let $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix}$ belong to H . By the One-Step Subgroup Test it suffices to show that $a - a' + b - b' + c - c' + d - d' = 0$. This follows from $a + b + c + d = 0 = a' + b' + c' + d'$. If 0 is replaced by 1, H is not a subgroup since it does not contain the identity.
66. Say $\det A = 2^m$ and $\det B = 2^n$. Then $\det(AB) = 2^{m+n}$ and $\det A^{-1} = 2^{-m}$.
67. If 2^a and $2^b \in K$, then $2^a(2^b)^{-1} = 2^{a-b} \in K$, since $a - b \in H$.
68. Let $f, g \in H$. Then $(f \cdot g^{-1})(2) = f(1)g^{-1}(2) = 1 \cdot 1 = 1$. The 2 can be replaced by any number.
69. $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ is not in H .
70. H is not closed.
71. If $a + bi$ and $c + di \in H$, then $(a + bi)(c + di)^{-1} = \frac{a+bi}{c+di} \frac{c-di}{c-di} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2} = (ac + bd) + (bc - ad)i$. Moreover,
 $(ac + bd)^2 + (bc - ad)^2 = a^2c^2 + 2acbd + b^2d^2 + b^2c^2 - 2bcad + a^2d^2$.
 Simplifying we obtain,
 $(a^2 + b^2)c^2 + (a^2 + b^2)d^2 = (a^2 + b^2)(c^2 + d^2) = 1 \cdot 1 = 1$. So, H is a subgroup. H is the unit circle in the complex plane.

72. Since G is Abelian the set is closed and therefore a subgroup.
 $|\langle a, b \rangle| \leq |a||b|$.
73. Since $ee = e$ is in $HZ(G)$ it is non-empty. Let h_1z_1 and h_2z_2 belong to $HZ(G)$. Then $h_1z_1(h_2z_2)^{-1} = h_1z_1z_2^{-1}h_2^{-1} = h_1h_2^{-1}z_1z_2^{-1} \in HZ(G)$.
74. Observe that if $a/b \in H$ and $c/d \in K$ where a, b, c, d are nonzero integers then $a = b(a/b) \in H$, $c = d(c/d) \in K$ and $ac \in H \cap K$.
75. First note that if $m/n \neq 0$ is element of H then $n(m/n) = m$ and $-m$ are also in H . By the Well Ordering Principle H has a least positive integer t . Since t is not in $K = \{2h \mid h \in H\}$ K is a nontrivial proper subgroup of H (see Example 5). Alternatively, one can use Exercise 74.
76. Suppose that G is a group of order $n > 2$ and H is a subgroup of G with $|H| = n - 1$. Let a be in G but a not in H and let b be in H and $b \neq e$. Then $ab \neq a$ and ab is not in H .
77. By the corollary of Theorem 0.2 there are integers s and t so that $1 = ms + nt$. Then $a^1 = a^{ms+nt} = a^{ms}a^{nt} = (a^m)^s(a^n)^t = (a^t)^n$.
78. Let $g \in G$, $g \neq e$. If $|g| = pm$, then $|g^m| = p$.

CHAPTER 4

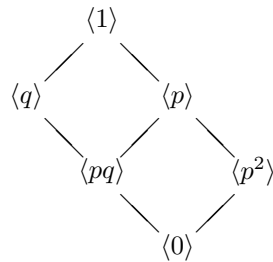
Cyclic Groups

1. For Z_6 , generators are 1 and 5; for Z_8 generators are 1, 3, 5, and 7; for Z_{20} generators are 1, 3, 7, 9, 11, 13, 17, and 19.
2. For $\langle a \rangle$, generators are a and a^5 ; for $\langle b \rangle$, generators are b , b^3 , b^5 , and b^7 ; for $\langle c \rangle$, generators are c , c^3 , c^7 , c^9 , c^{11} , c^{13} , c^{17} , c^{19} .
3. $\langle 20 \rangle = \{20, 10, 0\}$; $\langle 10 \rangle = \{10, 20, 0\}$
 $\langle a^{20} \rangle = \{a^{20}, a^{10}, a^0\}$; $\langle a^{10} \rangle = \{a^{10}, a^{20}, a^0\}$
4. $\langle 3 \rangle = \{3, 6, 9, 12, 15, 0\}$;
 $\langle 15 \rangle = \{15, 12, 9, 6, 3, 0\}$; $\langle a^3 \rangle = \{a^3, a^6, a^9, a^{12}, a^{15}, a^0\}$;
 $\langle a^{15} \rangle = \{a^{15}, a^{12}, a^9, a^6, a^3, a^0\}$.
5. $\langle 3 \rangle = \{3, 9, 7, 1\}$
 $\langle 7 \rangle = \{7, 9, 3, 1\}$
6. In any group, $\langle a \rangle = \langle a^{-1} \rangle$. See Exercise 11.
7. $U(8)$ or D_3 .
8. (a) All have order 5. (b) Both have order 3. (c) All have order 15.
9. Six subgroups; generators are the divisors of 20.
 Six subgroups; generators are a^k , where k is a divisor of 20.
10. $3 \cdot 1, 3 \cdot 3, 3 \cdot 5, 3 \cdot 7; a^3, (a^3)^3, (a^3)^5, (a^3)^7$.
11. By definition, $a^{-1} \in \langle a \rangle$. So, $\langle a^{-1} \rangle \subseteq \langle a \rangle$. By definition, $a = (a^{-1})^{-1} \in \langle a^{-1} \rangle$. So, $\langle a \rangle \subseteq \langle a^{-1} \rangle$.
12. $\langle 3 \rangle, \langle 3 \rangle; a^3, a^{-3}$.
13. Observe that $\langle 21 \rangle = \{0, 21, 18, 15, 12, 9, 6, 3\}$ and $\langle 10 \rangle = \{0, 10, 20, 6, 16, 2, 12, 22, 8, 18, 4, 14\}$. Since the intersection of two subgroups is a subgroup, according to the proof of Theorem 4.3, we can find a generator of the intersection by taking the smallest positive multiple of 1 that is in the intersection. So, $\langle 21 \rangle \cap \langle 10 \rangle = \langle 6 \rangle$.
 Similarly, $\langle a^{21} \rangle = \{e, a^{21}, a^{18}, a^{15}, a^{12}, a^9, a^6, a^3\}$ and $\langle a^{10} \rangle = \{e, a^{10}, a^{20}, a^6, a^{16}, a^2, a^{12}, a^{22}, a^8, a^{18}, a^4, a^{14}\}$. Then again by the proof of Theorem 4.3, we can find a generator of the intersection by taking the smallest positive power of a that is in the intersection. So, $\langle a^{21} \rangle \cap \langle a^{10} \rangle = \langle a^6 \rangle$.

For the case $\langle a^m \rangle \cap \langle a^n \rangle$, let $k = \text{lcm}(m, n)$. Write $k = ms$ and $k = nt$. Then $a^k = (a^m)^s \in \langle a^m \rangle$ and $a^k = (a^n)^t \in \langle a^n \rangle$. So, $\langle a^k \rangle \subseteq \langle a^m \rangle \cap \langle a^n \rangle$. Now let a^r be any element in $\langle a^m \rangle \cap \langle a^n \rangle$. Then r is a multiple of both m and n . It follows that r is a multiple of k (see Exercise 10 of Chapter 0). So, $a^r \in \langle a^k \rangle$.

14. 49. First note that the group is not infinite since an infinite cyclic group has infinitely many subgroups. Let $|G| = n$. Then 7 and $n/7$ are both divisors of n . If $n/7 \neq 7$, then G has at least 4 divisors. So, $n/7 = 7$. When 7 is replaced by p , $|G| = p^2$.
15. $|g|$ divides 12 is equivalent to $g^{12} = e$. So, if $a^{12} = e$ and $b^{12} = e$, then $(ab^{-1})^{12} = a^{12}(b^{12})^{-1} = ee^{-1} = e$. The same argument works when 12 is replaced by any integer (see Exercise 47 of Chapter 3).
16. $|a| = |a^2|$ if and only if $|a|$ is odd or infinite. To see this note that if $|a| = \infty$, then $|a^2|$ cannot be finite and if $|a| = n$, by Theorem 4.2 we have $n = |a^2| = n/\text{gcd}(n, 2)$ and therefore $\text{gcd}(n, 2) = 1$.
17. $|a^2| = |a^{12}|$ if and only if $|a| = \infty$ or $|a|$ is finite and $\text{gcd}(|a|, 2) = \text{gcd}(|a|, 12)$.
18. Both i and j are 0 or both are not 0.
19. 1 (the identity). To see this note that we can let the group be $\langle a \rangle$ where $|a|$ is infinite. If some element a^i has finite order n then $(a^i)^n = e$. But then $a^{in} = e$, which implies that a has finite order. This contradicts our assumption.
20. By Corollary 2 of Theorem 4.1 a nonidentity element of G must have order 5, 7 or 35. We may assume that G has no element of order 35. Since 34 is not a multiple of $\phi(5) = 4$, not all of the nonidentity elements can have order 5. Similarly, not all of them can have order 7. So, G has elements of orders both 5 and 7. Say, $|a| = 5$ and $|b| = 7$. Then, since $(ab)^5 = b^5 \neq e$ and $(ab)^7 = a^7 = a^2 \neq e$, we must have $|ab| = 35$, a contradiction.
21. **a.** $|a|$ divides 12. **b.** $|a|$ divides m . **c.** By Theorem 4.3, $|a| = 1, 2, 3, 4, 6, 8, 12$, or 24. If $|a| = 2$, then $a^8 = (a^2)^4 = e^4 = e$. A similar argument eliminates all other possibilities except 24.
22. Let $G = \{e, a, b\}$. Cancellation shows ab must be e . Thus $G = \{e, a, a^{-1}\}$.
23. Yes, by Theorem 4.3. The subgroups of Z are of the form $\langle n \rangle = \{0, \pm n, \pm 2n, \pm 3n, \dots\}$, for $n = 0, 1, 2, 3, \dots$. The subgroups of $\langle a \rangle$ are of the form $\langle a^n \rangle$ for $n = 0, 1, 2, 3, \dots$.
24. Certainly, $a \in C(a)$. Thus, $\langle a \rangle \subseteq C(a)$.

25. D_n has n reflections each of which has order 2. D_n also has n rotations that form a cyclic group of order n . So, according to Theorem 4.4, there are $\phi(d)$ rotations of order d in D_n . If n is odd, there are no rotations of order 2. If n is even, there is $\phi(2) = 1$ rotation of order 2. (Namely, R_{180} .) So, when n is odd D_n has n elements of order 2; when n is even, D_n has $n + 1$ elements of order 2.
26. 1 and -1 are the only generators of Z . Suppose that a^k generates $\langle a \rangle$. Then there is an integer t so that $(a^k)^t = a$. By Theorem 4.1, we conclude that $kt = 1$. So, $k = \pm 1$.
27. See Example 15 of Chapter 2.
28. The case that $i = -j$ is Exercise 4 of Chapter 3. If $\langle a^i \rangle = \langle a^j \rangle$, then $a^i = (a^j)^k = a^{jk}$ for some k . Since a has infinite order this means that $i = jk$ and therefore j divides i . Likewise, i divides j . So, $i = \pm j$.
29. 1000000, 3000000, 5000000, 7000000. By Theorem 4.3, $\langle 1000000 \rangle$ is the unique subgroup of order 8, and only those on the list are generators; $a^{1000000}, a^{3000000}, a^{5000000}, a^{7000000}$. By Theorem 4.3, $\langle a^{1000000} \rangle$ is the unique subgroup of order 8, and only those on the list are generators.
30. Let a be any nonidentity element of G . Since $\langle a \rangle$ is a subgroup of G other than $\{e\}$ it must be G . $\langle a \rangle$ is not infinite because $\langle a^2 \rangle$ would be a subgroup of G other than $\{e\}$ or G . By Theorem 4.3 the only divisors of $|a|$ are 1 and itself. So, $|a|$ is prime.
31. Let $G = \{a_1, a_2, \dots, a_k\}$. Now let $|a_i| = n_i$ and $n = n_1 n_2 \dots n_k$. Then $a_i^n = e$ for all i since n is a multiple of n_i .
- 32.



33.

$$\begin{array}{c}
\langle p^{n-n} \rangle \\
\vdots \\
\langle p^{n-3} \rangle \\
| \\
\langle p^{n-2} \rangle \\
| \\
\langle p^{n-1} \rangle \\
| \\
\langle 0 \rangle
\end{array}$$

34. First suppose that G is the union of proper subgroups. If G were cyclic, say $G = \langle a \rangle$, and G was the union of proper subgroups $H_1, H_2, \dots, H_n$, then a must be in one of H_i since the union contains every element. But if a belongs to H_i then $G = \langle a \rangle$ is a subgroup of the proper subgroup H_i . This is a contradiction. Now suppose that G is not cyclic. Then for every $g \in G$, $\langle g \rangle$ is a proper subgroup and $G = \cup_{g \in G} \langle g \rangle$.

35. Suppose that r is a generator of Q^+ . Since $\langle r \rangle = \langle r^{-1} \rangle$, we may assume that $r > 1$. Then there are positive integers m and n such that $r^m = 2$ and $r^n = 3$. Then $r^{mn} = (r^m)^n = 2^n$ and $r^{mn} = (r^n)^m = 3^m$. This implies that $2^n = 3^m$. But 2^n is even and 3^m is odd. This proves the group of nonzero rationals under multiplication is not cyclic for otherwise its subgroups would be cyclic.

36.

	4	8	12	16
4	16	12	8	4
8	12	4	16	8
12	8	16	4	12
16	4	8	12	16

The identity is 16. The group is generated by 8 and by 12.

37. For 6, use Z_{2^7} . For n , use $Z_{2^{n-1}}$.

38. $\text{lcm}(m, n)$

39. Suppose that $|ab| = n$. Then $(ab)^n = e$ implies that $b^n = a^{-n} \in \langle a \rangle$, which is finite. Thus $b^n = e$.

40. $|ab|$ could be any divisor of $\text{lcm}(|a|, |b|)$.

41. Since $\gcd(100, 98) = 2$ and $\gcd(100, 70) = 10$ we have $|a^{98}| = |a^2| = 50$ and $|a^{70}| = |a^{10}| = 10$.

42. Since FF' is a rotation other than the identity and the rotations of D_{21} form a cyclic subgroup of order 21, we know by Theorem 4.3 that $|FF'|$ is

a divisor of 21. Moreover, FF' cannot be the identity for then $FF' = FF$, which implies that $F' = F$. So, $|FF'| = 3, 7$ or 21 .

43. All divisors of 60.
44. Use the corollary to Theorem 4.4.
45. The argument given in the proof of the corollary to Theorem 4.4 shows that in an infinite group the number of elements of finite order n is a multiple of $\phi(n)$ or there is an infinite number of elements of order n .
46. Let $G = \langle g \rangle$ and $a = g^m$. Then $a^n = (g^m)^n = (g^n)^m = e^m = e$.
47. It follows from Example 15 in Chapter 2 and Example 12 in Chapter 0 that the group $H = \langle \cos(360^\circ/n) + i \sin(360^\circ/n) \rangle$ is a cyclic group of order n and every member of this group satisfies $x^n - 1 = 0$. Moreover, since every element of order n satisfies $x^n - 1 = 0$ and there can be at most n such elements, all complex numbers of order n are in H . Thus, by Theorem 4.4, C^* has exactly $\phi(n)$ elements of order n .
48. Clearly 0 is in H . If m and n are in H then m has the forms $8m_1$ and $10m_2$ and n has the forms $8n_1$ and $10n_2$. Then $m - n$ has the forms $8m_1 - 8n_1$ and $10m_2 - 10n_2$. So $m - n$ is in H . So, H is a subgroup of Z . If the condition is changed to “divisible by 8 or 10” H is not a subgroup since 8 and 10 would belong to H but $10 - 8 = 2$ would not.
49. Let $x \in Z(G)$ and $|x| = p$ where p is prime. Say $y \in G$ with $|y| = q$ where q is prime. Then $(xy)^{pq} = e$ and therefore $|xy| = 1, p$ or q . If $|xy| = 1$, then $x = y^{-1}$ and therefore $p = q$. If $|xy| = p$, then $e = (xy)^p = y^p$ and q divides p . Thus, $q = p$. A similar argument applies if $|xy| = q$.
50. If an infinite group had only a finite number of subgroups then it would have only a finite number of cyclic subgroups. If each of these cyclic subgroups is finite then the group would be finite since every element is in the cyclic subgroup generated by itself. So the group contains at least one infinite cyclic subgroup, call it $\langle a \rangle$. Then the subgroups $\langle a \rangle, \langle a^2 \rangle, \langle a^3 \rangle, \dots$ are distinct subgroups since the least positive power of a in $\langle a^i \rangle$ is a^i and a^i is not the least positive power of a in any subgroup $\langle a^j \rangle$ for $j \neq i$.
51. An infinite cyclic group does not have element of prime order. A finite cyclic group can have only one subgroup for each divisor of its order. A subgroup of order p has exactly $p - 1$ elements of order p . Another element of order p would give another subgroup of order p .
52. $2; 4; a^3, a^5, a^7$.
53. $1 \cdot 4, 3 \cdot 4, 7 \cdot 4, 9 \cdot 4; x^4, (x^4)^3, (x^4)^7, (x^4)^9$.
54. In group, the number of elements order d is divisible by $\phi(d)$ or there are infinitely many elements of order d .

55. D_{33} has 33 reflections each of which has order 2 and 33 rotations that form a cyclic group. So, according to Theorem 4.4, for each divisor d of 33 there are $\phi(d)$ rotations of order d in D_n . This gives one element of order 1; $\phi(3) = 2$ elements of order 3; $\phi(11) = 10$ elements of order 11; and $\phi(33) = 20$ elements of order 33.
56. Since $U(25) = 20$, by Corollary 1 of Theorem 4.2 we know that $|2|$ must divide 20. So, $|2| = 1, 2, 4, 5, 10$, or 20. But $2^{10} \neq 1$ implies that $|2| \neq 1, 2, 5$ or 10 and $2^4 \neq 1$ implies that $|2| \neq 4$.
57. Let $|\langle a \rangle| = 4$ and $|\langle b \rangle| = 5$. Since $(ab)^{20} = (a^4)^5(b^5)^4 = e \cdot e = e$ we know that $|ab|$ divides 20. Noting that $(ab)^4 = b^4 \neq e$ we know that $|ab| \neq 1, 2$ or 4. Likewise, $(ab)^{10} = a^2 \neq e$ implies that $|ab| \neq 5$ or 10. So, $|ab| = 20$. Then, by Theorem 4.3, $\langle ab \rangle$ has subgroups of orders 1, 2, 4, 5, 10 and 20. In general, if an Abelian group contains cyclic subgroups of order m and n where m and n are relatively prime, then it contains subgroups of order d for each divisor d of mn .
58. 1, 2, 3, 12. In general, if an Abelian group contains cyclic subgroups of order m and n , then it contains subgroups of order d for each divisor d of the least common multiple of m and n .
59. Say a and b are distinct elements of order 2. If a and b commute, then ab is a third element of order 2. If a and b do not commute, then aba is a third element of order 2.
60. 12
61. By Exercise 34 of Chapter 3, $\langle a \rangle \cap \langle b \rangle$ is a subgroup. Also, $\langle a \rangle \cap \langle b \rangle \subseteq \langle a \rangle$ and $\langle a \rangle \cap \langle b \rangle \subseteq \langle b \rangle$. So, by Theorem 4.3, $|\langle a \rangle \cap \langle b \rangle|$ is a common divisor of 10 and 21. Thus, $|\langle a \rangle \cap \langle b \rangle| = 1$ and therefore $\langle a \rangle \cap \langle b \rangle = \{e\}$.
62. Mimic Exercise 53.
63. $|\langle a \rangle \cap \langle b \rangle|$ must divide both 24 and 10. So, $|\langle a \rangle \cap \langle b \rangle| = 1$ or 2.
64. From Theorem 4.4 we know that a finite cyclic group has at most 1 element of order 2. Now observe that $2^{n-1} - 1$ and $2^n - 1$ have order 2.
65. Note that among the integers from 1 to p^n the p^{n-1} integers $p, 2p, \dots, p^{n-1}p$ are exactly the ones not relatively prime to p .
66. First note that if k is a generator then so is $-k$. Thus it suffices to show that $k \neq -k$. But $k = -k$ implies that $2k = 0$ so that $n = |k| = 1$ or 2.
67. Observe that $|a^5| = 12$ implies that $e = (a^5)^{12} = a^{60}$ so $|a|$ divides 60. Since $\langle a^5 \rangle \subseteq \langle a \rangle$ we know that $|\langle a \rangle|$ is divisible by 12. So, $|\langle a \rangle| = 12$ or 60. If $|a^4| = 12$, then $|a|$ divides 48. Since $\langle a^4 \rangle \subseteq \langle a \rangle$ we know that $|\langle a \rangle|$ is divisible by 12. So, $|\langle a \rangle| = 12, 24$, or 48. But $|a| = 12$ implies $|a^4| = 3$ and $|a| = 24$ implies $|a^4| = 6$. So, $|a| = 48$.

68. By Theorem 4.3, it suffices to find necessary and sufficient conditions so that $|x^r|$ divides $|x^s|$. By Theorem 4.2, we obtain $\gcd(n, s)$ divides $\gcd(n, r)$.
69. $\gcd(48, 21) = 3$; $\gcd(48, 14) = 2$; $\gcd(48, 18) = 6$.
70. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$.
71. Since $(ab)^{80} = (a^5)^{16}(b^{16})^5 = ee = e$ we know that $|ab|$ divides 80. Thus $|ab| = 1, 2, 4, 8, 16, 5, 10, 20, 40$, or 80. If $|ab|$ is 1, 2, 4, 8 or 16 then $e = (ab)^{16} = a^{16}b^{16} = a$, which is false. If $|ab|$ is 5, 10, 20, or 40 then $e = (ab)^{40} = a^{40}b^{40} = b^8$, which is false. So $|ab| = 80$.
72. By Exercise 34 of Chapter 3, $\langle a \rangle \cap \langle b \rangle$ is a subgroup $\langle a \rangle$ and $\langle b \rangle$. So, $|\langle a \rangle \cap \langle b \rangle|$ divides 12 and 22. It follows that $|\langle a \rangle \cap \langle b \rangle| = 1$ or 2 and since $\langle a \rangle \cap \langle b \rangle \neq \{e\}$ we have that $|\langle a \rangle \cap \langle b \rangle| = 2$. Because $\langle a^6 \rangle$ is the only subgroup of $\langle a \rangle$ of order 2 and $\langle b^{11} \rangle$ is the only subgroup of $\langle b \rangle$ of order 2, we have $\langle a \rangle \cap \langle b \rangle = \langle a^6 \rangle = \langle b^{11} \rangle$ and therefore $a^6 = b^{11}$.
73. $\phi(81) = 27 \cdot 2 = 54$; $\phi(60) = \phi(4)\phi(3)\phi(5) = 2 \cdot 2 \cdot 4 = 16$;
 $\phi(105) = \phi(3) \cdot \phi(5) \cdot \phi(7) = 2 \cdot 4 \cdot 6 = 48$.
74. Write $n = 2^k m$ where $k \geq 1$ and m is odd. Then
 $\phi(2n) = \phi(2^{k+1}m) = \phi(2^{k+1})\phi(m) = 2^k\phi(m) = 2 \cdot 2^{k-1}\phi(m) =$
 $2\phi(2^k)\phi(m) = 2\phi(2^k m) = 2\phi(n)$.
75. Since m and n are relatively prime, it suffices to show both m and n divide k . By Corollary 2 of Theorem 4.1, it is enough to show that $a^k = e$. Note that $a^k \in \langle a \rangle \cap \langle b \rangle$, and since $\langle a \rangle \cap \langle b \rangle$ is a subgroup of both $\langle a \rangle$ and $\langle b \rangle$, we know that $|\langle a \rangle \cap \langle b \rangle|$ must divide both $|\langle a \rangle|$ and $|\langle b \rangle|$. Thus, $|\langle a \rangle \cap \langle b \rangle| = 1$. For an example let a be any non-identity group element of finite order n and let $b = a^{-1}$. Then $a^n = b^n$ but n^2 does not divide n . Or let x be a group element of order 4 and let $a = x$, $b = x^2$ and $k = 4$.
76. Note that n and $n^2 - 2$ are distinct elements of order 2 and appeal to Theorem 4.4.
77. First note that $x \neq e$. If $x^3 = x^5$, then $x^2 = e$. By Corollary 2 Theorem 4.1 and Theorem 4.3 we then have $|x|$ divides both 2 and 15. Thus $|x| = 1$ and $x = e$. If $x^3 = x^9$, then $x^6 = e$ and therefore $|x|$ divides 6 and 15. This implies that $|x| = 3$. Then $|x^{13}| = |x(x^3)^4| = |x| = 3$. If $x^5 = x^9$, then $x^4 = e$ and $|x|$ divides both 4 and 15, and therefore $x = e$.

CHAPTER 5

Permutation Groups

1. **a.** $\alpha^{-1} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{bmatrix}$
b. $\beta\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 6 & 2 & 3 & 4 & 5 \end{bmatrix}$
c. $\alpha\beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 2 & 1 & 5 & 3 & 4 \end{bmatrix}$
2. $\alpha = (12345)(678) = (15)(14)(13)(12)(68)(67); \beta = (23847)(56) = (27)(24)(28)(23)(56); \alpha\beta = (12485736) = (16)(13)(17)(15)(18)(14)(12).$
3. **a.** $(15)(234)$ **b.** $(124)(35)(6)$ **c.** (1423)
4. 2; 3; 5; k .
5. **a.** By Theorem 5.3 the order is $\text{lcm}(3,3) = 3$.
b. By Theorem 5.3 the order is $\text{lcm}(3,4) = 12$.
c. By Theorem 5.3 the order is $\text{lcm}(3,2) = 6$.
d. By Theorem 5.3 the order is $\text{lcm}(3,6) = 6$.
e. $|(1235)(24567)| = |(124)(3567)| = \text{lcm}(3,4) = 12$.
f. $|(345)(245)| = |(25)(34)| = \text{lcm}(2,2) = 2$.
6. 6; 12
7. By Theorem 5.3 the order is $\text{lcm}(4,6) = 12$.
8. $(135) = (15)(13)$ even; $(1356) = (16)(15)(13)$ odd; $(13567) = (17)(16)(15)(13)$ even; $(12)(134)(152) = (12)(14)(13)(12)(15)$ odd; $(1243)(3521) = (13)(14)(12)(31)(32)(35)$ even.
9. We find the orders by looking at the possible products of disjoint cycle structures arranged by longest lengths left to right and denote an n -cycle by $(\underline{n})$.
 $(\underline{6})$ has order 6 and is odd;
 $(\underline{5})(\underline{1})$ has order 5 and is even;
 $(\underline{4})(\underline{2})$ has order 4 and is even;
 $(\underline{4})(\underline{1})(\underline{1})$ has order 4 and is odd;
 $(\underline{3})(\underline{3})$ has order 3 and is even;
 $(\underline{3})(\underline{2})(\underline{1})$ has order 6 and is odd;
 $(\underline{3})(\underline{1})(\underline{1})(\underline{1})$ has order 3 and is even;
 $(\underline{2})(\underline{2})(\underline{2})$ has order 2 and is odd;

- (2)(2)(1)(1) has order 2 and is even;
 (2)(1)(1)(1)(1) has order 2 and is odd.
 So, for S_6 , the possible orders are 1, 2, 3, 4, 5, 6; for A_6 the possible orders are 1, 2, 3, 4, 5. We see from the cycle structure of S_7 shown in Example 4 that in A_7 the possible orders are 1, 2, 3, 4, 5, 6, 7.
10. $|(123)(45678)| = 15$
11. $(12345)(678)(9,10)(11,12)$ is in A_{12} and has order 30.
12. Say $S = \{s_1, \dots, s_n\}$ and ϕ is one-to-one from S to S . Then $\phi(s_1), \dots, \phi(s_n)$ are all distinct and all in S so $\phi(S) = S$. On the other hand, if $\phi(s_i) = \phi(s_j)$ for some $i \neq j$, then $\phi(S)$ has at most $n - 1$ members. The mapping from Z to Z that takes x to $2x$ is one-to-one but not onto.
13. To prove that α is 1-1 assume $\alpha(x_1) = \alpha(x_2)$. Then $x_1 = \alpha(\alpha(x_1)) = \alpha(\alpha(x_2)) = x_2$. To prove that α is onto note that for any s in S , we have $\alpha(\alpha(s)) = s$.
14. First observe that α is odd and β is even. Then all odd powers of α are odd and all powers of β are even. So, the product has three odds and 2 evens, which is odd. Compare with Exercise 20 of Chapter 5.
15. An n -cycle is even when n is odd since we can write it as a product of $n - 1$ 2-cycles by successively pairing up the first element of the cycle with each of the other cycle elements starting from the last element of the cycle and working towards the front. The same process shows that when n is even we get an odd permutation.
16. If $\alpha_1, \alpha_2, \dots, \alpha_n$ are 2-cycles and $\alpha = \alpha_1 \cdots \alpha_n$ then $\alpha^{-1} = \alpha_n \alpha_{n-1} \cdots \alpha_2 \alpha_1$.
17. An even number of two cycles followed by an even number of two cycles gives an even number of two cycles in all. So the Finite Subgroup Test is verified.
18. even; odd.
19. Suppose that α can be written as a product on m 2-cycles and β can be written as product of n 2-cycles. Then juxtaposing these 2-cycles we can write $\alpha\beta$ as a product of $m + n$ 2-cycles. Now observe that $m + n$ is even if and only if m and n are even or both odd.
20.
$$\begin{array}{lll} (+1) & \cdot & (+1) = (+1) \\ \text{even} & \cdot & \text{even} = \text{even} \end{array} \qquad \begin{array}{lll} (-1) & \cdot & (-1) = +1 \\ \text{odd} & \cdot & \text{odd} = \text{even} \end{array}$$

$$\begin{array}{lll} (+1) & \cdot & (-1) = (-1) \\ \text{even} & \cdot & \text{odd} = \text{odd} \end{array} \qquad \begin{array}{lll} (-1) & \cdot & (+1) = (-1) \\ \text{odd} & \cdot & \text{even} = \text{odd} \end{array}$$

21. The number of odd cycles in the product is even.
22. $a_n a_{n-1} \cdots a_2 a_1$
23. If all members of H are even we are done. So, suppose that H has at least one odd permutation σ . For each odd permutation β in H observe that $\sigma\beta$ is even and, by cancellation, different β s give different $\sigma\beta$ s. Thus, there are at least as many even permutations as there are odd ones. Conversely, for each even permutation β in H observe that $\sigma\beta$ is odd and, by cancellation, different β s give different $\sigma\beta$ s. Thus, there are at least as many odd permutations as there are even ones.
24. By Exercise 23, either every element of H is even or half are even and half are odd. In the latter case, H would have even order.
25. The identity is even; the set is not closed.
26. If α can be written as the product of m 2-cycles and β can be written as a product of n 2-cycles, then $\alpha^{-1}\beta^{-1}\alpha\beta$ can be written as the product of $n + m + n + m = 2(m + n)$ 2-cycles.
27. $(8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1)/(2 \cdot 2 \cdot 2 \cdot 2 \cdot 4!)$
28. $(7 \cdot 6 \cdot 5 \cdot 4 \cdot 3)/5$
29. An odd permutation of order 4 must be of the form $(a_1 a_2 a_3 a_4)$. There are 6 choices for a_1 , 5 for a_2 , 4 for a_3 , and 3 for a_4 . This gives $6 \cdot 5 \cdot 4 \cdot 3$ choices. But since for each of these choices the cycles $(a_1 a_2 a_3 a_4) = (a_2 a_3 a_4 a_1) = (a_3 a_4 a_1 a_2) = (a_4 a_1 a_2 a_3)$ give the same group element we must divide $6 \cdot 5 \cdot 4 \cdot 3$ by 4 to obtain 90. An even permutation of order 4 must be of the form $(a_1 a_2 a_3 a_4)(a_5 a_6)$. As before, there are 90 choices $(a_1 a_2 a_3 a_4)$. Since $(a_5 a_6) = (a_6 a_5)$ there are 90 elements of the form $(a_1 a_2 a_3 a_4)(a_5 a_6)$. This gives 180 elements of order 4 in S_6 .
A permutation in S_6 of order 2 has three possible disjoint cycle forms: $(a_1 a_2)$, $(a_1 a_2)(a_3 a_4)$ and $(a_1 a_2)(a_3 a_4)(a_5 a_6)$. For $(a_1 a_2)$ there are $6 \cdot 5/2 = 15$ distinct elements; for $(a_1 a_2)(a_3 a_4)$ there are $6 \cdot 5 \cdot 4 \cdot 3$ choices for the four entries but we must divide by $2 \cdot 2 \cdot 2$ since $(a_1 a_2) = (a_2 a_1)$, $(a_3 a_4) = (a_4 a_3)$ and $(a_1 a_2)(a_3 a_4) = (a_4 a_3)(a_1 a_2)$. This gives 45 distinct elements. For $(a_1 a_2)(a_3 a_4)(a_5 a_6)$ there are $6!$ choices for the six entries but we must divide by $2 \cdot 2 \cdot 2 \cdot 3!$ since each of the three 2-cycles can be written 2 ways and the three 2-cycles can be permuted $3!$ ways. This gives 15 elements. So, the total number of elements of order 2 is 75.
30. Any product of 3-cycles is even whereas (1234) is odd. In general, no odd permutation can be written as a product of 3-cycles.

31. Since $\beta^{28} = (\beta^4)^7 = \epsilon$, we know that $|\beta|$ divides 28. But $\beta^4 \neq \epsilon$ so $|\beta| \neq 1, 2$, or 4. If $|\beta| = 14$, then β written in disjoint cycle form would need at least one 7-cycle and one 2-cycle. But that requires at least 9 symbols and we have only 7. Likewise, $|\beta| = 28$ requires at least one 7-cycle and one 4-cycle. So, $|\beta| = 7$. Thus, $\beta = \beta^8 = (\beta^4)^2 = (2457136)$. In S_9 , $\beta = (2457136)$ or $\beta = (2457136)(89)$.
32. Observe that $\beta = (123)(145) = (14523)$ so that $\beta^{99} = \beta^4 = \beta^{-1} = (13254)$.
33. Since $|(a_1a_2a_3a_4)(a_5a_6)| = 4$ such an x would have order 8. But the elements in S_{10} of order 8 are 8-cycles or the disjoint product of 8-cycle and a 2-cycle. In both cases the square of such an element is the product of two 4-cycles.
34. If α and β are disjoint 2-cycles, then $|\alpha\beta| = \text{lcm}(2,2) = 2$. If α and β have exactly one symbol in common we can write $\alpha = (ab)$ and $\beta = (ac)$. Then $\alpha\beta = (ab)(ac) = (acb)$ and $|\alpha\beta| = 3$.
35. Let $\alpha, \beta \in \text{stab}(a)$. Then $(\alpha\beta)(a) = \alpha(\beta(a)) = \alpha(a) = a$. Also, $\alpha(a) = a$ implies $\alpha^{-1}(\alpha(a)) = \alpha^{-1}(a)$ or $a = \alpha^{-1}(a)$.
36. Since $|\beta| = 21$, we have $n = 16$.
37. Since $\alpha^m = (1, 3, 5, 7, 9)^m(2, 4, 6)^m(8, 10)^m$ and the result is a 5-cycle we deduce that $(2, 4, 6)^m = \epsilon$ and $(8, 10)^m = \epsilon$. So, 3 and 2 divide m . Since $(1, 3, 5, 7, 9)^m \neq \epsilon$ we know that 5 does not divide m . Thus, we can say that m is a multiple of 6 but not a multiple of 30.
38. Let $\beta, \gamma \in H$. Then $(\beta\gamma)(1) = \beta(\gamma(1)) = \beta(1) = 1$;
 $(\beta\gamma)(3) = \beta(\gamma(3)) = \beta(3) = 3$. So, by Theorem 3.3, H is a subgroup.
 $|H| = 6$. The proof is valid for all $n \geq 3$. In the general case,
 $|H| = (n-2)!$. When S_n is replaced by A_n , $|H| = (n-2)!/2$.
39. $\langle (1234) \rangle; \{(1), (12), (34), (12)(34)\}$
40. Let $\alpha = (12)$ and $\beta = (13)$.
41. Let $\alpha = (123)$ and $\beta = (145)$.
42. $R_0 = (1)(2)(3); R_{120} = (123); R_{240} = (132); (12); (13); (23)$.
43. Observe that (12) and (123) belong to S_n for all $n \geq 3$ and they do not commute.
44. Observe that $(123)(124) \neq (124)(123)$.
45. In disjoint cycle form elements of H are exactly those for which 1 and 2 appear as the cycles $(1)(2)$ or (12) . The identity is in H since it has the form $(1)(2)$. Let α and β belong to H . If for both α and β we have the cycles $(1)(2)$, then for $\alpha\beta$ we have $(1)(2)$. If for both α and β we have the

cycle (12), then for $\alpha\beta$ we have (1)(2). If for one of α and β we have the cycles (1)(2) and for the other (12), then for $\alpha\beta$ we have (12). So, by the Finite Subgroup Test, H is a subgroup. To find $|H|$ observe that in matrix form we have 2 choices (1 or 2) for the image of 1, the second entry must be the choice of 1 or 2 not used as the image of 1, and $(n-2)!$ choices for the remaining $n-2$ images. So, $|H| = 2(n-2)!$

46. For any permutation α in S_7 , α^2 is even whereas (1234) is odd. For the second part we can take $x = (1432), (1432)(567), (1432)(576)$
47. Theorem 5.2 shows that disjoint cycles commute. For the other half, assume that $a = c$. Since $(ab) \neq (cd)$, we know $b \neq d$. If $(ab)(cd) = (cd)(ab)$ then $(ab)(cd) = (ab)(ad) = (adb)$ and $(cd)(ab) = (ad)(ab) = (abd)$. This means that a maps to both b and d , which are distinct. But permutations are 1-1 mappings.
48. Say β can be written with m 2-cycles and α with n 2-cycles. Then $\beta^{-1}\alpha\beta$ can be written with $2m + n$ 2-cycles.
49. R_0, R_{180}, H, V .
50. 216° rotation; reflection about the axis joining vertex 1 to the midpoint of the opposite side.
51. The permutation corresponding to the rotation of $360/n$ degrees, $(1, 2, \dots, n)$, is an even permutation so all rotations are even.
52. $\alpha_1\alpha_2\alpha_3$ is odd and ϵ is even. The product of an odd number of 2-cycles cannot ϵ .
53. Cycle decomposition shows any nonidentity element of A_5 is a 5-cycle, a 3-cycle, or a product of a pair of disjoint 2-cycles. Then, observe there are $(5 \cdot 4 \cdot 3 \cdot 2 \cdot 1)/5 = 24$ group elements of the form $(abcde)$, $(5 \cdot 4 \cdot 3)/3 = 20$ group elements of the form (abc) and $(5 \cdot 4 \cdot 3 \cdot 2)/8 = 15$ group elements of the form $(ab)(cd)$. In this last case we must divide by 8 because there are 8 ways to write the same group element $(ab)(cd) = (ba)(cd) = (ab)(dc) = (ba)(dc) = (cd)(ab) = (cd)(ba) = (dc)(ab) = (dc)(ba)$.
54. One possibility for a cyclic subgroup is $\langle (1234)(5678) \rangle$. One possibility for a noncyclic subgroup is $\{(1), (12)(34), (56)(78), (12)(34)(56)(78)\}$.
55. If α has odd order n and α is an odd permutation then $\epsilon = \alpha^n$ would be an odd permutation.
56. Using the notation in Table 5.1, α_2, α_3 , and α_4 have order 2; $\alpha_5, \alpha_6, \dots, \alpha_{12}$ have order 3. The orders of the elements divide the order of the group.

57. Any element from A_n is expressible as a product of an even number of 2-cycles. For each pair of 2-cycles there are two cases. One is that they share an element in common $(ab)(ac)$ and the other is that they are disjoint $(ab)(cd)$. Now observe that $(ab)(ac) = (abc)$ and $(ab)(cd) = (cbd)(acb)$.
58. Suppose $\alpha \neq \varepsilon$ and $\alpha \in Z(S_n)$. Write α in disjoint cycle form $(a_1 a_2 \dots) \dots$ where $a_1 \neq a_2$. Let a_3 be different from a_1 and a_2 and let $\beta = (a_1)(a_2 a_3)$. Then $(\alpha\beta)(a_1) = a_2$ while $(\beta\alpha)(a_1) = a_3$.
59. That $a * \sigma(b) \neq b * \sigma(a)$ is done by examining all cases. To prove the general case, observe that $\sigma^i(a) * \sigma^{i+1}(b) \neq \sigma^i(b) * \sigma^{i+1}(a)$ can be written in the form $\sigma^i(a) * \sigma(\sigma^i(b)) \neq \sigma^i(b) * \sigma(\sigma^i(a))$, which is the case already done. If a transposition were not detected, then $\sigma(a_1) * \dots * \sigma^i(a_i) * \sigma^{i+1}(a_{i+1}) * \dots * \sigma^n(a_n) = \sigma(a_1) * \dots * \sigma^i(a_{i+1}) * \sigma^{i+1}(a_i) * \dots * \sigma^n(a_n)$, which implies $\sigma^i(a_i) * \sigma^{i+1}(a_{i+1}) = \sigma^i(a_{i+1}) * \sigma^{i+1}(a_i)$.
60. 5
61. By Theorem 5.4 it is enough to prove that every 2-cycle can be expressed as a product of elements of the form $(1k)$. To this end observe that if $a \neq 1, b \neq 1$, then $(ab) = (1a)(1b)(1a)$.
62. Let α denote the permutation of positions induced by a shuffling. Label the positions ace to king as 1 through 13. We are given that $\alpha^2 = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \\ 8 & 12 & 6 & 7 & 9 & 11 & 13 & 4 & 2 & 1 & 10 & 3 & 5 \end{bmatrix} = (1, 8, 4, 7, 13, 5, 9, 2, 12, 3, 6, 11, 10)$.
- Since $|\alpha^2| = 13$ we know that $|\alpha| = 13$ or 26. But S_{13} has no elements of order 26. So, $|\alpha| = 13$. Thus, $\alpha = \alpha^{14} = (1, 2, 8, 12, 4, 3, 7, 6, 13, 11, 5, 10, 9)$.
63. By case-by-case analysis H is a subgroup for $n = 1, 2, 3$ and 4. For $n \geq 5$, observe that $(12)(34)$ and $(12)(35)$ belong to H but their product does not.
64. In Exercise 35 let G be A_5 . Then $\text{stab}(1)$ is the subgroup of A_5 consisting of the 24 even permutations of the set $\{2, 3, 4, 5\}$. Similarly, $\text{stab}(2)$, $\text{stab}(3)$, $\text{stab}(4)$, $\text{stab}(5)$ are subgroups of order 24.
65. The product of an element from $Z(A_4)$ of order 2 and an element of A_4 of order 3 would have order 6. But A_4 has no element of order 6.
66. Since $|\alpha|$ is the least common multiple of disjoint cycles lengths of α let the distinct disjoint cycle lengths be α be $n_1, n_2, \dots, n_k$. Because α be $n_1, n_2, \dots, n_k$ are distinct integers between 1 and n each of them appears as a term in $n!$.
67. TAAKTPKSTOOPEDN

68. ADVANCE WHEN READY